**Exp no:9** Implement Dielectric free space boundary condition in Electric Field and find the relative permittivity of the different medium.

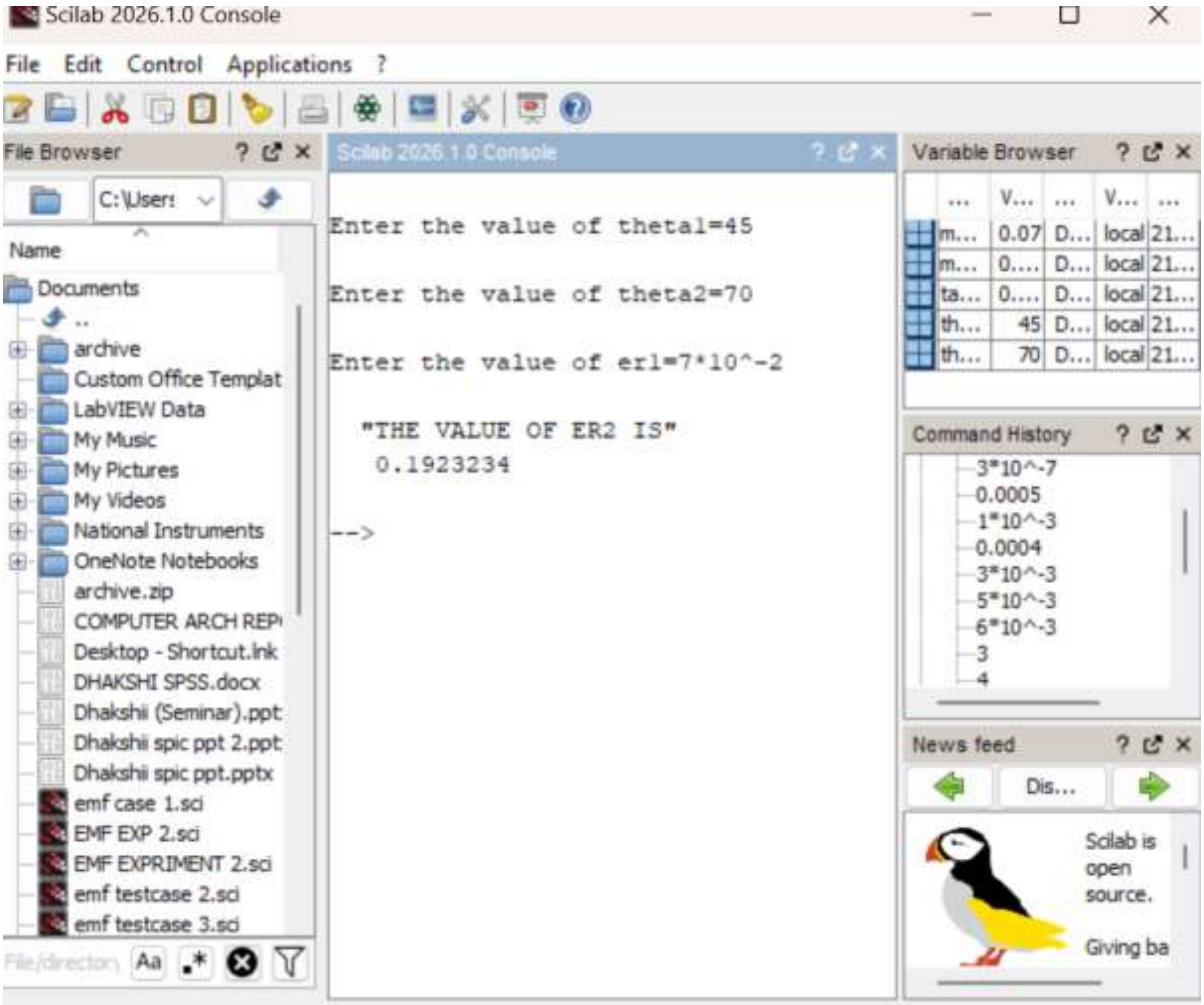
**Aim:**

To write program for Dielectric - free space boundary condition in Electric field using SCILAB.

**Software required:**

•SCILAB version 6.0.1

**Partical calucation:**



exp 9.sci (C:\Users\dhakshinya velan\Documents\exp 9.sci) - SciNotes

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EMF EXPRIMENT 2.sci | EXP6.sci | exp 7.sci | exp 8.sci | exp 9.sci

```
clc;
clear;

theta_1=input("Enter the value of theta1=");
theta_2=input("Enter the value of theta2=");
medium_1=input("Enter the value of er1=");

tan_theta= tand(theta_1)/tand(theta_2);
medium_2= medium_1/tan_theta;
disp('THE VALUE OF ER2 IS',[medium_2]);
```

**Theoritical calucation:**

9 = $\theta_1 = 45^0$

$\theta_2 = 70^0$

$\varepsilon r_1 = 7\times10^{-2}$

$\varepsilon_{r2} = \dfrac{\varepsilon_{r1}}{\text{tan-theta}}$

$= \dfrac{.007}{.36396}$

$= .01923$

$\text{tan-theta} = \dfrac{\tan}{\tan}$

$= \dfrac{1}{2.7475}$

$= 0.36396$